AI MANTHAN 2.0 · PS9 CROP HEALTH MAPPING

FasalNetra

Satellite finds the weak patch. The farmer confirms it. Drone Didi sprays only that patch.

फसलनेत्र · the crop's eye, for soybean farmers of Malwa

THE PROBLEM

Farmers find disease late, then spray the whole field for a problem in one corner.

ONE YELLOW PATCH, THREE CAUSES

Virus, water stress or nutrient shortage. Each needs a different fix, and a wrong guess wastes the spray.

MANDSAUR 2025

Yellow mosaic outbreak; compensation survey ordered on **1 Sept 2025**.

INDIA SOYBEAN OUTPUT

125.82 → **105.36** lakh t (2024 → 2025)

WHAT WE LEARNED FROM ICAR-IISR

We consulted Dr. Savita Kolhe, ICAR-IISR Indore. Three points shaped the design.

- **Warn about 15 days early.** After symptoms show, it is too late. So we predict first.
- **Drone imaging is unreliable:** rotor wind shakes the plants. So the drone only sprays; the farmer's phone collects the evidence.
- **Farmers trust the weekly ICAR advisory.** So every spray recommendation comes from it, with the source read out.

HOW IT WORKS · THE FARMER ONLY PRESSES 1

From a weather warning to a spray of only the mapped zone

1

Risk engine scores every field daily from weather, crop stage and advisory

2

Hindi voice call warns the farmer and asks yes/no questions on the keypad

3

SMS link opens a voice-guided page; no app, no typing

4

Grid walk to the satellite-flagged 10 m cells; photo of leaf top and underside

5

Evidence card: every signal votes into one confidence score

6

High: advice + zone-only spray. **Otherwise:** Drone Didi checks first

PROOF 1 · MANDSAUR 2025 BACKTEST

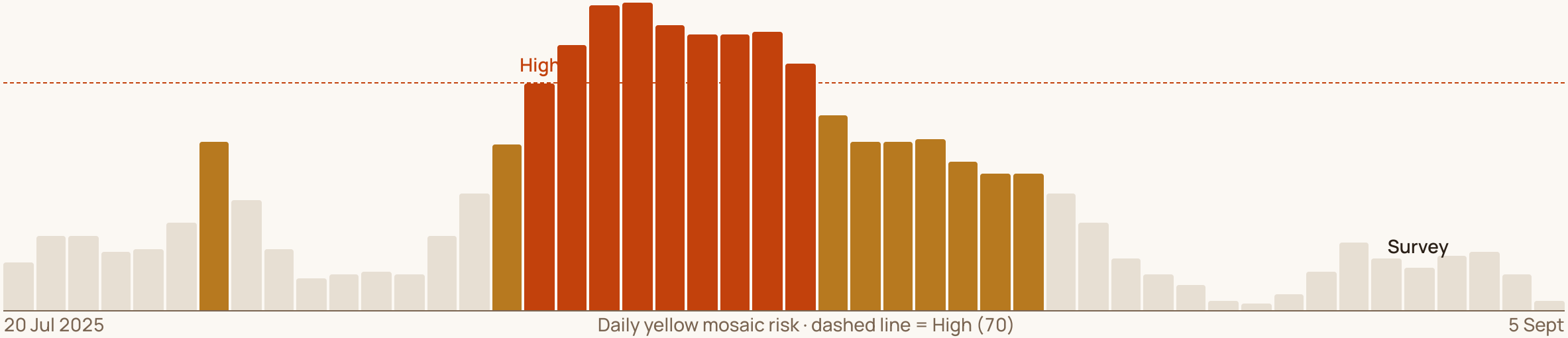
27 days

warning before the 1 Sept outbreak survey. Risk turned High on **5 Aug** for all **8 of 8** sowing dates tested (25 Jun to 16 Jul).

CONTROL SEASON

2024: 0 High days with the same rule and field, so it does not alarm every monsoon.

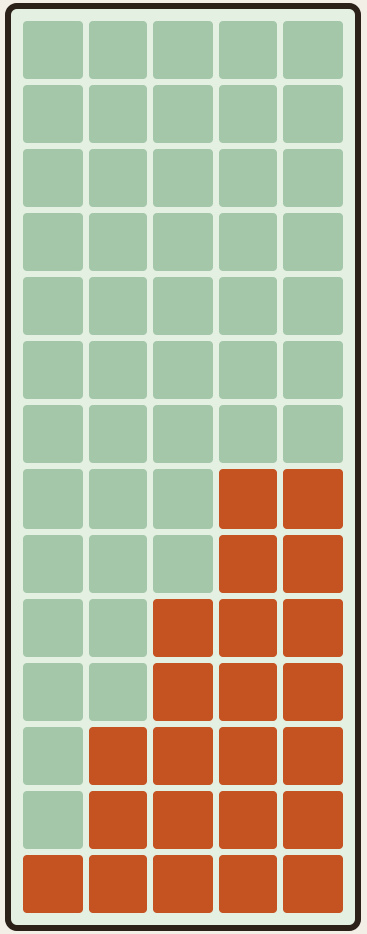
Honest limits: one outbreak event; rule thresholds are provisional until checked against the MP yellow mosaic paper.



PROOF 2 · SENTINEL-2 AS A SCOUT, NOT A JUDGE

Each 10 m cell is compared with its own field and its own history.

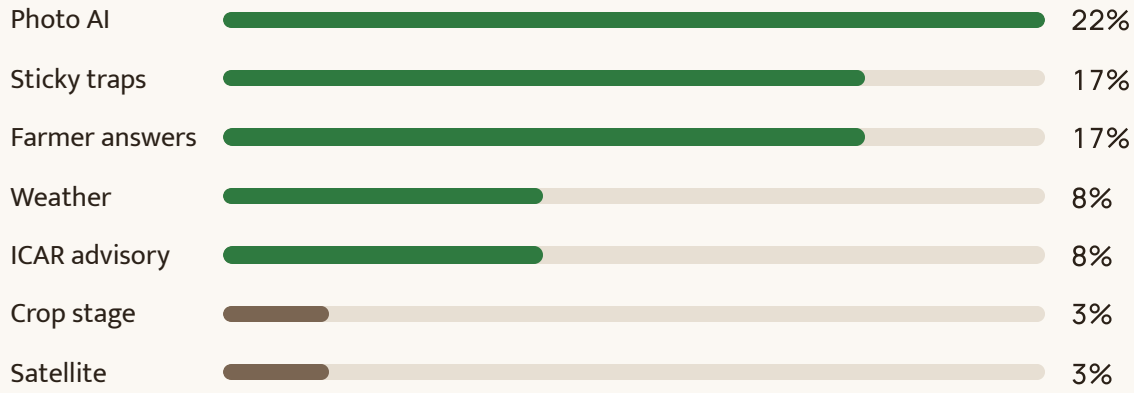
- Demo field, scene **1 Aug 2025**: **27 of 69** cells flagged, one compact patch of **0.67 acre**.
- Cells on the field edge are never flagged: they mix in the neighbour's land.
- Monsoon clouds left only **4** clear scenes in Jul to Sep 2025. That is why the farmer's walk confirms what the satellite suggests.



TRUST · EVERY SIGNAL VOTES

No single signal decides. Doubt sends a human.

- Photos + weather + satellite only: Medium 66% → Drone Didi checks first
- Add the farmer's "yes, whiteflies" and trap counts: High 77% → zone-only spray
- Patch in a low spot: spray blocked, waterlogging checked first
- Always "shak" (suspected), never a certain diagnosis



USE OF GENERATIVE AI

AI where it is strong; rules and sources where farmers need certainty.

VISION LLM

Closed label list, few-shot Indian reference photos, symptoms written before the label. Test and calibration harness for SoyNet and MH-SoyaHealthVision; accuracy {{photo_accuracy}} once run.

COST AWARE

Fast model first, stronger model only when unsure. Retries only on transient errors. Budget tracked per session.

VOICE IN HINDI

One decision per prompt, 9 repeats, 0 reaches a human. Money is spent only after a key press.

NO INVENTED DOSES

The model picks a cause; the dose comes only from the ICAR-NSRI advisory table.

DRONE DIDI CONSOLE · WHERE, WHAT, WHEN

A Namo Drone Didi gets a job she can act on at a glance.

- **Where:** the zone and a lawnmower route that never leaves it, as an ArduPilot .waypoints file
- **What:** product, ml for this zone, litres of water, safety steps
- **When:** next morning's hours with no rain and wind under 10 km/h

DEMO RUN, 1.58 ACRE FIELD

0.83 acre

not sprayed · **42 ml** chemical saved · **12** passes, **4.9** min flight

SOCIAL IMPACT · SUSTAINABILITY

Less chemical, earlier action, more work for women drone operators.

FARMER

Warned before symptoms, sprays only the sick patch, gets dated geotagged proof for crop insurance.

DRONE DIDI

More farms per battery, because each job covers only the zone. Builds on the existing Namo Drone Didi scheme.

WHO PAYS

Drone Didi SHGs and FPOs, input companies, state agriculture departments, crop insurers.

PILOT

1 Drone Didi, **10** farmers, **1** kharif season in Mandsaur.

CROSS-DOMAIN INTEGRATION · BUILT AND TESTED

Remote sensing, agronomy, telephony, generative AI and drones in one loop.

DATA

Sentinel-2 (Planetary Computer), Open-Meteo, Copernicus DEM 30 m, ICAR-NSRI advisories, Indian soybean photo sets.

STACK

FastAPI, SQLite, React + Leaflet, Twilio voice and SMS, Gemini vision, ArduPilot mission files.

QUALITY

41 automated tests, **81%** coverage, Hindi and English locale checks, lint clean.

FASALNETRA

The farmer only presses 1.

Live now: we call a judge's phone, walk the field link together, and send Drone Didi to spray only the patch.